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Season 3

Case Study No.3

Team Details

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Reliable source of navigation for off-highway vehicles in environments that may deter line of sight for GNSS satellites due to buildings & trees

- Reliable navigation of off-highway vehicles in GNSS-challenged environments remains a critical bottleneck for safety, productivity, and autonomy. Urban canyons, dense vegetation, and mining sites frequently degrade satellite visibility, causing intermittent positioning and unbounded errors when relying on GNSS-centric solutions. This abstract proposes a **self-reliant localization approach** that maintains positioning continuity and integrity even under prolonged GNSS degradation, without heavy dependence on external infrastructure.
- The solution introduces an **adaptive confidence-based localization stack** that fuses GNSS, inertial sensing, wheel odometry, and onboard perception while continuously evaluating sensor reliability in real time. Instead of assuming equal sensor trust, the system dynamically re-weights inputs based on motion context, environmental observability, and fault indicators. When GNSS quality drops, the framework seamlessly transitions to dead-reckoning and perception-anchored localization with bounded drift, ensuring graceful degradation rather than abrupt failure.
- Designed specifically for off-highway constraints, the architecture emphasizes edge-efficient computation, minimal calibration overhead, and robustness to vibration, dust, thermal stress, and wheel slip. Short-term local mapping enables spatial consistency in rapidly changing sites without requiring high-maintenance global maps. The approach targets high availability, predictable error growth, and operational scalability across diverse vehicle platforms such as loaders, dumpers, and tractors.
- By prioritizing adaptability, integrity monitoring, and infrastructure independence, the proposed solution delivers a practical and scalable navigation foundation suitable for real-world off-highway operations, addressing both current deployment challenges and future autonomy requirements.